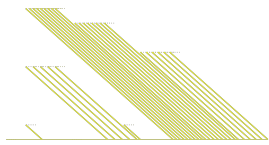


Chapitre 5 : Les tests d'hypothèses

Marie VAUGOYEAU (contact@marievaugoyeau.fr)

Table des matières

1	Avant de commencer	2
2	Test bilatérale, unilatéral droit ou gauche	2
3	Test de Shapiro-Wild	5
4	Loi Khi-2	8
4.1	Test d'adéquation du khi-2	10
4.2	Test d'indépendance du Khi-2	10
4.3	Test d'homogénéité	13



1 Avant de commencer

Ce chapitre reprends les codes du chapitre 5 sur *les tests d'hypothèses* du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2^{ème} édition).

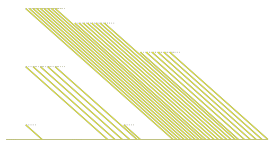
```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2     4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr       1.2.2
— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```

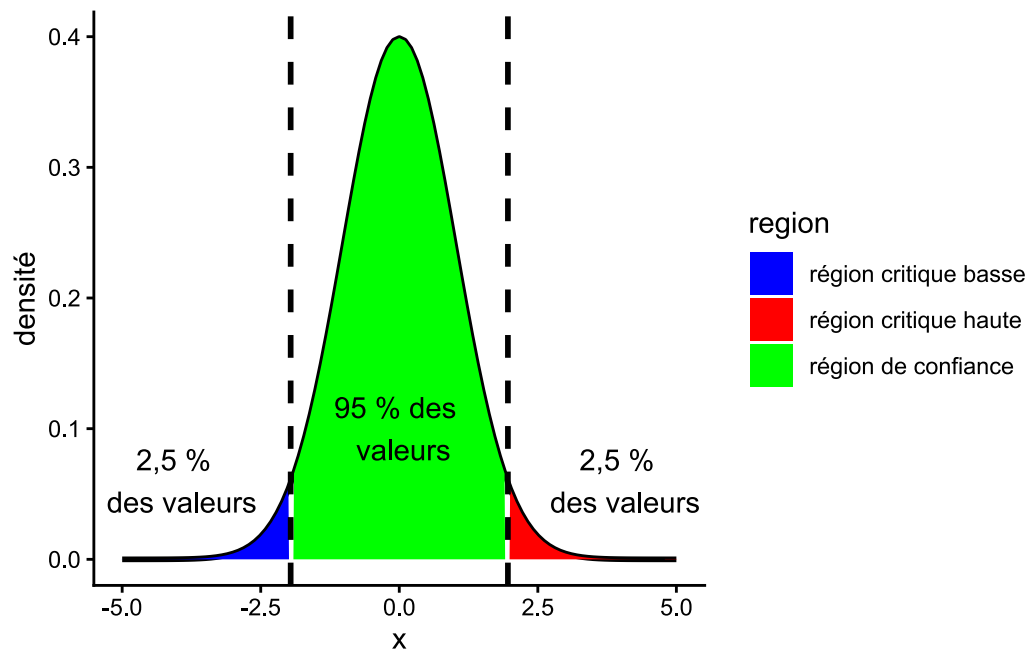
2 Test bilatérale, unilatéral droit ou gauche

```
# choix de l'intervalle de représentation
intervalle <- seq(-5, 5, 0.1)

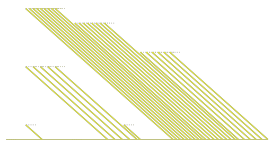
# test bilatéral
tibble(
  x = intervalle,
  "densité" = dnorm(intervalle, mean = 0, sd = 1),
  region =
    case_when(
      x < qnorm(0.025) ~ "région critique basse",
      x < qnorm(0.975) ~ "région de confiance",
      TRUE ~ "région critique haute"
    )
) |>
ggplot() +
  aes(x = x, y = `densité`) +
  geom_line(linewidth = 1) +
  geom_area(aes(fill = region)) +
  scale_fill_manual(values = c("blue", "red", "green")) +
  geom_vline(xintercept = qnorm(0.025), linewidth = 1, linetype = "dashed") +
  geom_vline(xintercept = qnorm(0.975), linewidth = 1, linetype = "dashed") +
```



```
annotate("text", x = -4, y = 0.06, label = "2,5 % \n des valeurs") +  
annotate("text", x = 4, y = 0.06, label = "2,5 % \n des valeurs") +  
annotate("text", x = 0, y = 0.1, label = "95 % des \n valeurs") +  
theme_classic()
```

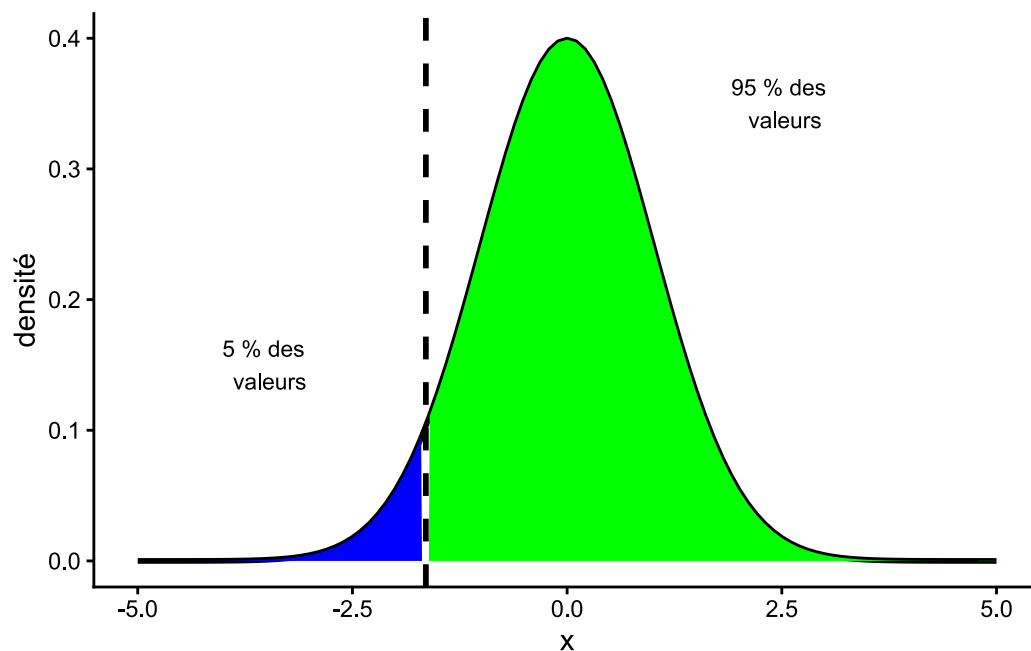


```
ggsave("img/05Rl01N.png", width = 14, height = 6, units = "cm")  
  
# test unilatéral à gauche  
tibble(  
  x = intervalle,  
  "densité" = dnorm(intervalle, mean = 0, sd = 1),  
  region =  
    case_when(  
      x < qnorm(0.05) ~ "région critique basse",  
      TRUE ~ "région de confiance"  
    )  
) |>  
  ggplot() +  
  aes(x = x, y = `densité`) +  
  geom_line(size = 1) +  
  geom_area(aes(fill = region)) +  
  scale_fill_manual(values = c("blue", "green")) +  
  geom_vline(xintercept = qnorm(0.05), linewidth = 1, linetype = "dashed") +
```

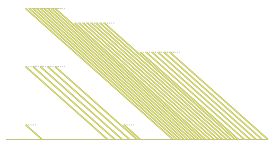


```
annotate("text", x = -3.5, y = 0.15, label = "5 % des \n valeurs", size = 3) +  
annotate("text", x = 2.5, y = 0.35, label = "95 % des \n valeurs", size = 3) +  
theme_classic() +  
theme(legend.position = "none")
```

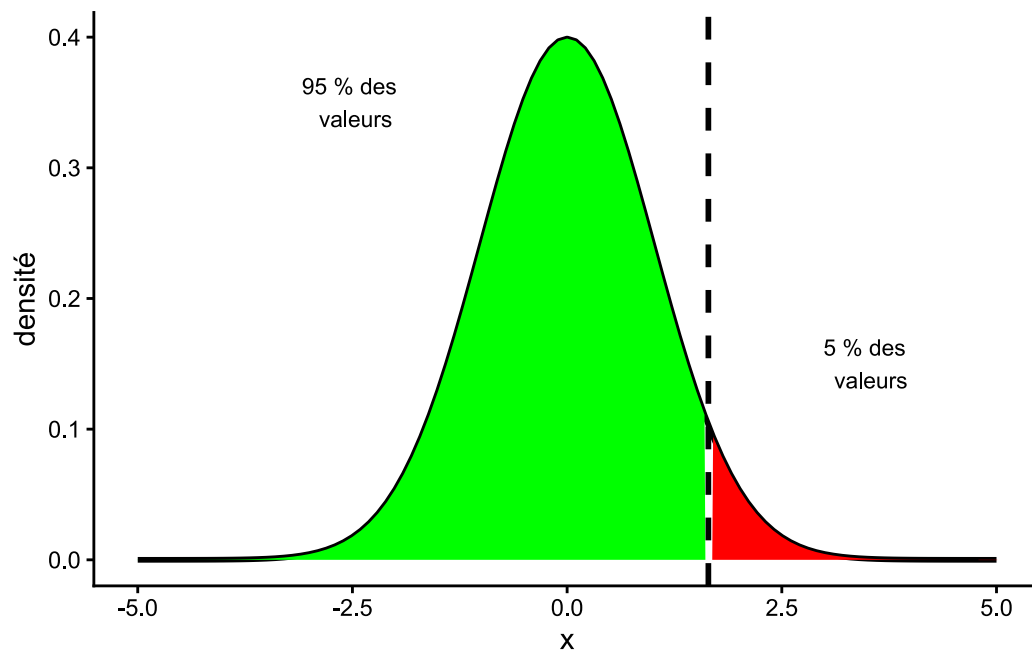
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



```
ggsave("img/05Rl01bisN.png", width = 6, height = 6, units = "cm")  
  
# test unilatéral à droite  
tibble(  
  x = intervalle,  
  "densité" = dnorm(intervalle, mean = 0, sd = 1),  
  region =  
    case_when(  
      x > qnorm(0.95) ~ "région critique haute",  
      TRUE ~ "région de confiance"  
    )  
) |>  
  ggplot() +  
  
  aes(x = x, y = `densité`) +
```



```
geom_line(size = 1) +  
geom_area(aes(fill = region)) +  
scale_fill_manual(values = c("red", "green")) +  
geom_vline(xintercept = qnorm(0.95), linewidth = 1, linetype = "dashed") +  
annotate("text", x = 3.5, y = 0.15, label = "5 % des \n valeurs", size = 3) +  
annotate("text", x = -2.5, y = 0.35, label = "95 % des \n valeurs", size =  
3) +  
theme_classic() +  
theme(legend.position = "none")
```



```
ggsave("img/05Rl01terN.png", width = 6, height = 6, units = "cm")
```

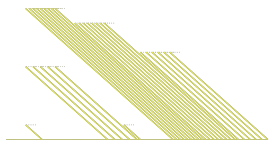
3 Test de Shapiro-Wild

limite à 5000

```
shapiro.test(rnorm(10000, mean = 5, sd = 3))
```

```
Error in `shapiro.test()`:  
! la taille de l'échantillon doit être comprise entre 3 et 5000
```

exemple réalisation test de Shapiro

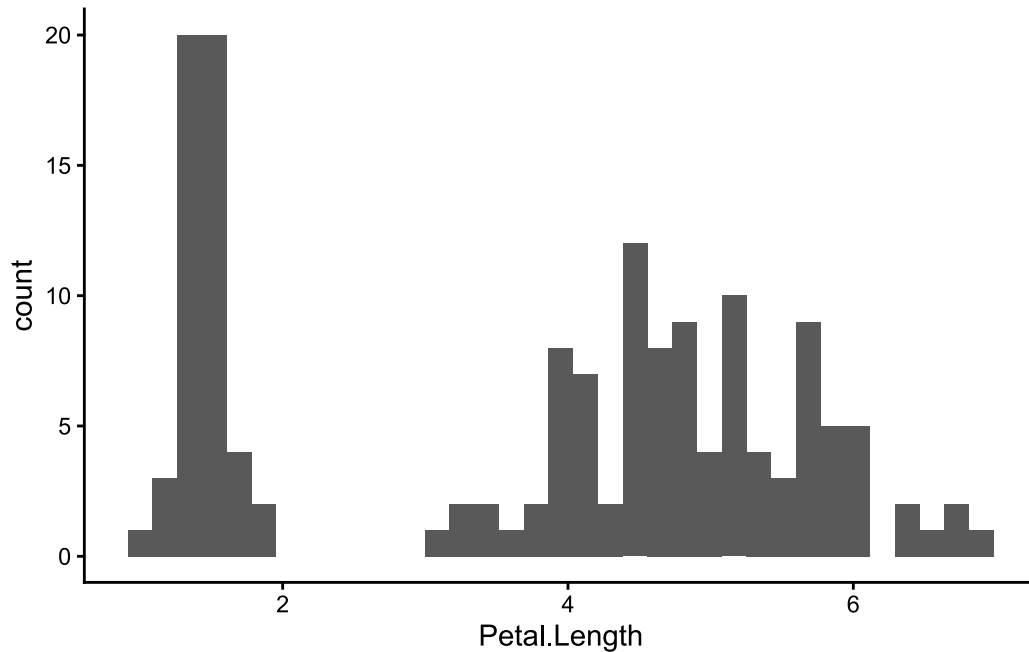


```
shapiro.test(iris$Petal.Length)
```

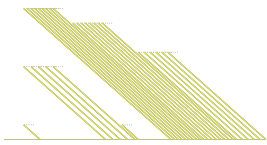
Shapiro-Wilk normality test

```
data: iris$Petal.Length  
W = 0.87627, p-value = 7.412e-10
```

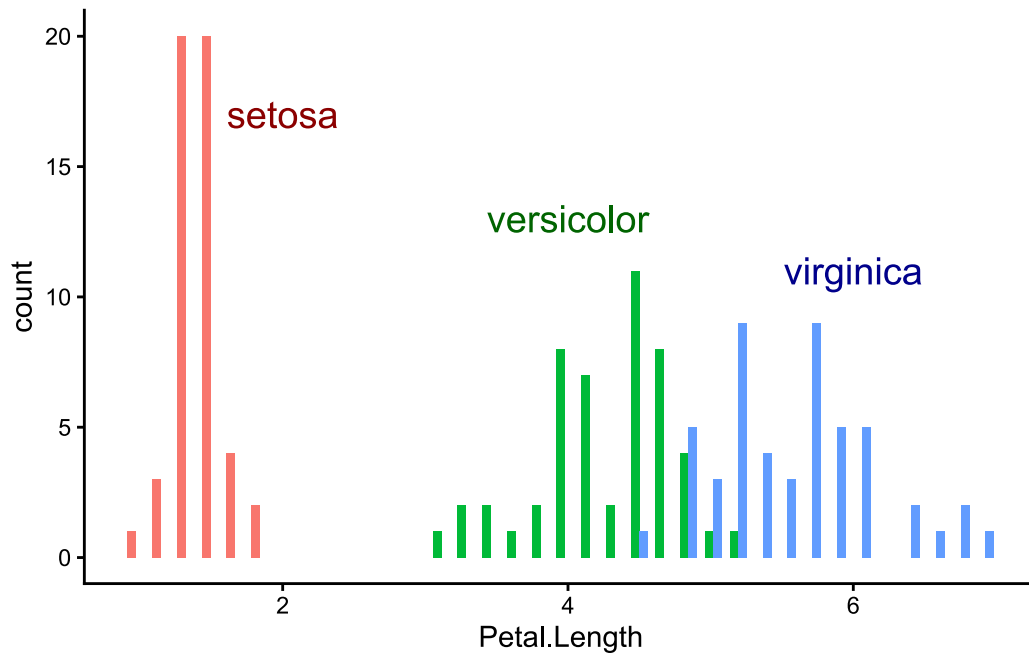
```
ggplot(iris) +  
  aes(x = Petal.Length) +  
  geom_histogram(bins = 35) +  
  theme_classic()
```



```
ggplot(iris) +  
  aes(x = Petal.Length, fill = Species) +  
  geom_histogram(bins = 35, position = "dodge") +  
  annotate("text", x = 2, y = 17, size = 5, label = "setosa", color = "darkred") +  
  annotate("text", x = 4, y = 13, size = 5, label = "versicolor", color =  
"darkgreen") +  
  annotate("text", x = 6, y = 11, size = 5, label = "virginica", color =  
"darkblue") +
```



```
theme_classic() +  
theme(legend.position = "none")
```



```
ggsave("img/05Rl04N.png", width = 14, height = 10, units = "cm")
```

```
shapiro.test((iris |> filter(Species == "setosa"))$Petal.Length)
```

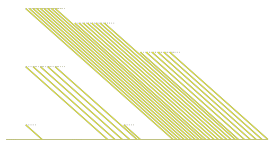
Shapiro-Wilk normality test

```
data: (filter(iris, Species == "setosa"))$Petal.Length  
W = 0.95498, p-value = 0.05481
```

```
shapiro.test((iris |> filter(Species != "setosa"))$Petal.Length)
```

Shapiro-Wilk normality test

```
data: (filter(iris, Species != "setosa"))$Petal.Length  
W = 0.99099, p-value = 0.7445
```



```
shapiro.test((iris |> filter(Species == "versicolor"))$Petal.Length)
```

Shapiro-Wilk normality test

```
data: (filter(iris, Species == "versicolor"))$Petal.Length  
W = 0.966, p-value = 0.1585
```

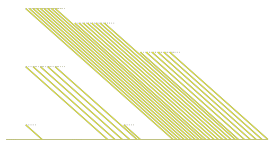
```
shapiro.test((iris |> filter(Species == "virginica"))$Petal.Length)
```

Shapiro-Wilk normality test

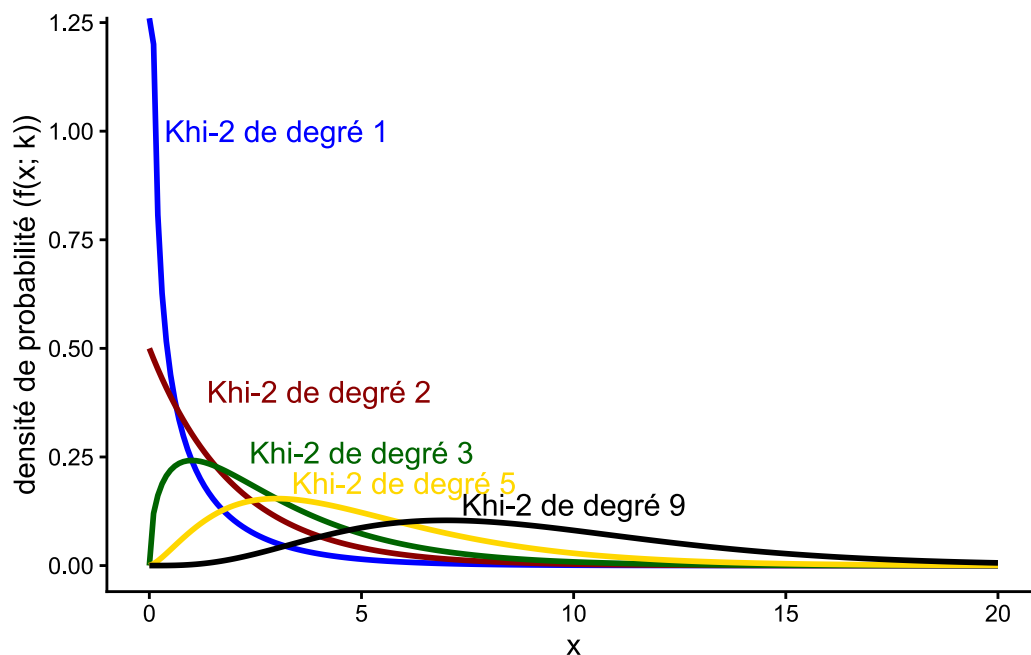
```
data: (filter(iris, Species == "virginica"))$Petal.Length  
W = 0.96219, p-value = 0.1098
```

4 Loi Khi-2

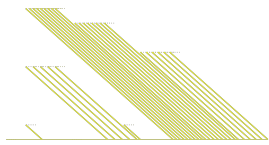
```
# choix de l'intervalle de représentation  
intervalle <- seq(0, 20, 0.1)  
  
# loi khi-deux de paramètres k égale à 1, 2, 3, 5 ou 9  
tibble(  
  loi_khi_deux =  
    rep(  
      c(  
        "k = 1",  
        "k = 2",  
        "k = 3",  
        "k = 5",  
        "k = 9"  
      ),  
      each = length(intervalle)  
    ),  
  x = rep(intervalle, 5),  
  densite =  
    c(  
      dchisq(intervalle, df = 1),  
      dchisq(intervalle, df = 2),  
      dchisq(intervalle, df = 3),  
      dchisq(intervalle, df = 5),  
      dchisq(intervalle, df = 9)
```

```
dchisq(intervalle, df = 9)
)
) |>
  ggplot() +
  aes(x = x, y = densite, color = loi_khi_deux) +
  geom_line(size = 1) +
  theme_classic() +
  scale_color_manual(values = c("blue", "darkred", "darkgreen", "gold", "black"))
+
  annotate("text", x = 3, y = 1, size = 4, color = "blue", label = "Khi-2 de
degré 1") +
  annotate("text", x = 4, y = 0.4, size = 4, color = "darkred", label = "Khi-2
de degré 2") +
  annotate("text", x = 5, y = 0.26, size = 4, color = "darkgreen", label = "Khi-2
de degré 3") +
  annotate("text", x = 6, y = 0.19, size = 4, color = "gold", label = "Khi-2 de
degré 5") +
  annotate("text", x = 10, y = 0.14, size = 4, color = "black", label = "Khi-2
de degré 9") +
  ylab("densité de probabilité (f(x; k))") +
  theme(legend.position = "none")
```



```
ggsave("img/05Rl06N.png", width = 14, height = 10, units = "cm")
```



4.1 Test d'adéquation du khi-2

```
chisq.test(  
  x = c(86, 111, 107, 91, 105, 100),  
  p = rep(1/6, 6)  
)
```

Chi-squared test for given probabilities

```
data: c(86, 111, 107, 91, 105, 100)  
X-squared = 4.72, df = 5, p-value = 0.451
```

```
chisq.test(  
  x = c(86, 111, 107, 91, 105, 100)  
)
```

Chi-squared test for given probabilities

```
data: c(86, 111, 107, 91, 105, 100)  
X-squared = 4.72, df = 5, p-value = 0.451
```

```
chisq.test(  
  x = c(4, 2)  
)
```

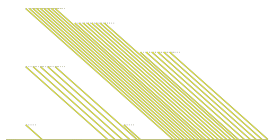
Warning in `chisq.test(x = c(4, 2))`: L'approximation du Chi-2 est peut-être incorrecte

Chi-squared test for given probabilities

```
data: c(4, 2)  
X-squared = 0.66667, df = 1, p-value = 0.4142
```

4.2 Test d'indépendance du Khi-2

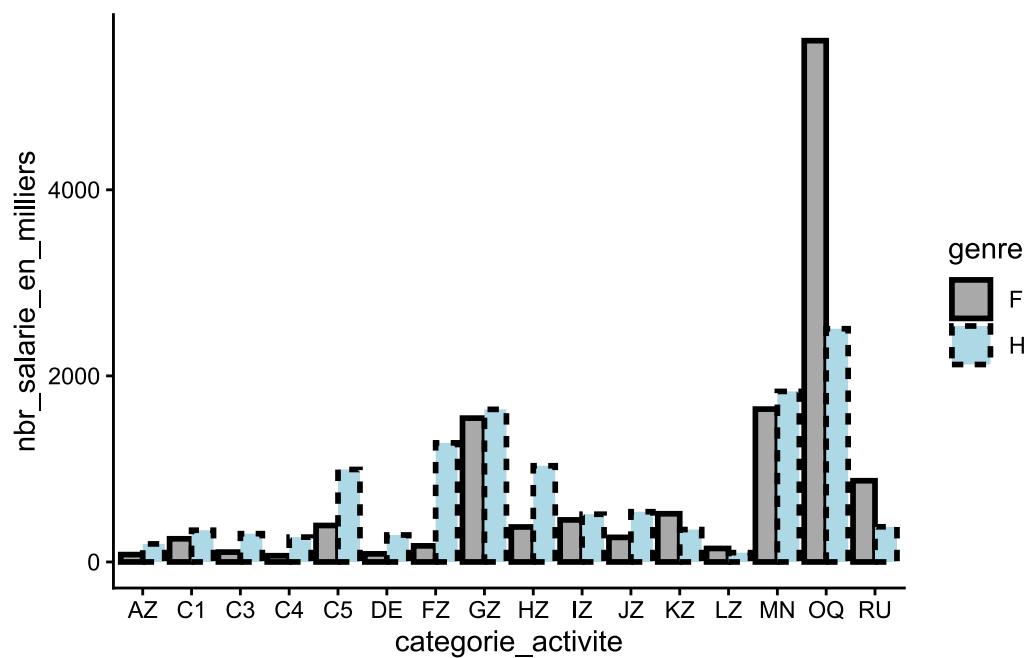
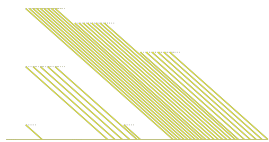
création du tableau de données



```
table_genre_naf <-  
  tibble(  
    categorie_activite = rep(c("AZ", "C1", "C3", "C4", "C5", "DE", "FZ", "GZ",  
"HZ", "IZ", "JZ", "KZ", "LZ", "MN", "OQ", "RU"), 2),  
    genre = rep(c("F", "H"), each = 16),  
    nbr_salarie_en_milliers = c(79, 249, 106, 67, 392, 87, 173, 1546, 376, 452,  
264, 518, 145, 1643, 5603, 873, 194, 341, 304, 267, 994, 290, 1280, 1640, 1034,  
513, 540, 348, 100, 1833, 2507, 378)  
  )
```

création du graphique

```
ggplot(table_genre_naf) +  
  aes(  
    x = categorie_activite,  
    y = nbr_salarie_en_milliers,  
    fill = genre,  
    color = genre,  
    linetype = genre  
  ) +  
  geom_bar(  
    stat = "identity",  
    position = "dodge",  
    linewidth = 1,  
    color = "black"  
  ) +  
  scale_fill_manual(values = c("darkgrey", "lightblue")) +  
  theme_classic()
```

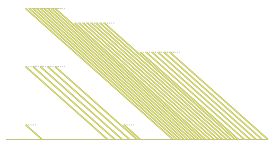


```
ggsave("img/05Rl10N.png", width = 14, height = 7, units = "cm")
```

```
data.frame(
  Femmes = c(79, 249, 106, 67, 392, 87, 173, 1546, 376, 452, 264, 518, 145, 1643, 5603, 873),
  Hommes = c(194, 341, 304, 267, 994, 290, 1280, 1640, 1034, 513, 540, 348, 100, 1833, 2507, 378),
  row.names = c("AZ", "C1", "C3", "C4", "C5", "DE", "FZ", "GZ", "HZ", "IZ", "JZ", "KZ", "LZ", "MN", "OQ", "RU")
) |>
  chisq.test()
```

Pearson's Chi-squared test

```
data: data.frame(Femmes = c(79, 249, 106, 67, 392, 87, 173, 1546, 376, 452, 264, 518, 145, 1643, 5603, 873), Hommes = c(194, 341, 304, 267, 994, 290, 1280, 1640, 1034, 513, 540, 348, 100, 1833, 2507, 378), row.names = c("AZ", "C1", "C3", "C4", "C5", "DE", "FZ", "GZ", "HZ", "IZ", "JZ", "KZ", "LZ", "MN", "OQ", "RU"))
X-squared = 3330.6, df = 15, p-value < 2.2e-16
```



4.3 Test d'homogénéité

```
library(rvest)
```

Attachement du package : 'rvest'

L'objet suivant est masqué depuis 'package:readr':

```
guess_encoding
```

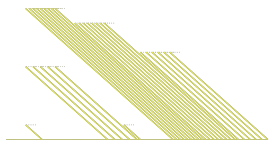
```
html <- read_html("https://www.insee.fr/fr/statistiques/4768237")
```

```
prop_cadre_genre <- html |>  
  html_element("table") |>  
  html_table(dec = ",") |>  
  as_tibble(.name_repair = make.names)
```

```
prop_cadre_genre
```

```
# A tibble: 38 × 4  
      X Femmes Hommes Ensemble  
  <int> <dbl> <dbl> <dbl>  
1  1982     4   10.3    7.8  
2  1983    4.4    11    8.3  
3  1984    4.5   11.3    8.5  
4  1985    4.6   11.2    8.5  
5  1986    5.2   11.7     9  
6  1987    5.5    12    9.2  
7  1988    5.9   12.7    9.8  
8  1989    5.9   12.4    9.7  
9  1990    6.7   13.2   10.4  
10 1991    7.3   13.8    11  
# i 28 more rows
```

```
# exemple pour la sixième table  
(read_html("https://www.insee.fr/fr/statistiques/4768237") |>  
  html_elements("table"))[[3]] |>  
  html_table(dec = ",")
```



```
# A tibble: 8 × 4
  ` ` Femmes Hommes Ensemble
  <chr> <dbl> <dbl> <dbl>
1 Professions libérales 47.4 52.6 100
2 Cadres de la fonction publique 50.4 49.6 100
3 Professeurs, professions scientifiques 55.5 44.5 100
4 Professions de l'information, des arts et des spectacl... 44.6 55.4 100
5 Cadres administratifs et commerciaux d'entreprise 49.3 50.7 100
6 Ingénieurs et cadres techniques d'entreprise 23.2 76.8 100
7 Ensemble des cadres et professions intellectuelles sup... 42.2 57.8 100
8 Ensemble des personnes en emploi 48.5 51.5 100
```

```
# test du KLhi-2 d'homogénéité
prop_cadre_genre |>
  rename(annee = X) |>
  filter(annee > 1999) |>
  select(c(Femmes, Hommes)) |>
  chisq.test()
```

Pearson's Chi-squared test

```
data: select(filter(rename(prop_cadre_genre, annee = X), annee > 1999),
c(Femmes, Hommes))
X-squared = 0.72221, df = 19, p-value = 1
```